

# SOURAV YADAV

Web3 Engineer | Solidity Developer | DeFi Protocol Engineer | Smart Contract Security Researcher

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📍 Madhya Pradesh, India

🔑 [GitHub](#) | [Sherlock](#) | [Code4rena](#) | [Cantina](#) | [Cyfrin](#) | [LeetCode](#)

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## PROFESSIONAL SUMMARY

Web3 engineer specializing in Solidity development, DeFi protocol engineering, and smart contract security, with hands-on experience building and auditing decentralized systems.

Audited **20+ DeFi protocols** across Sherlock, Code4rena, and Cantina, identifying **10+ validated vulnerabilities**. Built full-stack Web3 infrastructure including **pre-transaction security middleware**, **yield-generating payment commitment protocols**, and **multi-chain DEX routing systems**.

## TECHNICAL SKILLS

- **Smart Contracts:** Solidity, Foundry, Hardhat, Open Zeppelin, Proxy Patterns, Gas Optimization.
- **Security:** Fuzz Testing, Invariant Testing, Slither, Aderyn, Mythril, Tenderly, Exploit PoC Development.
- **DeFi:** AMMs, Uniswap V2/V3, Aave V3/V4, ERC4626 Vaults, Lending, Staking, Protocol Accounting, DEX Routing.
- **Frontend:** React.js, TypeScript, JavaScript (ESM), HTML5, CSS3, Tailwind CSS.
- **Tools & Platforms:** Git, GitHub, Linux, Vercel, Netlify.
- **Web3 / Backend:** Ethers.js, Viem, Wagmi, Web3.js, The Graph, Chainlink (CRE, Price Feed, Automation, Functions, VRF), Node.js, Express.js, RPCs, REST APIs, MySQL, PostgreSQL.
- **Languages:** Solidity, C++, JavaScript, TypeScript, Rust, Java (basic).
- **Networks:** Ethereum, Arbitrum, Base, Optimism, Polygon, BNB Chain, Anchor, Solana, Wasm.

## PROJECTS

### PreFlight — Pre-Transaction Security Middleware

[GitHub Repository](#)

*Solidity · Foundry · Chainlink CRE · React · Viem · Browser Extension · V2 Pools · ERC4626*

- Built a **browser-extension-based transaction security middleware** that intercepts user transactions before execution across DEX and ERC4626 vault interactions.
- Engineered hybrid risk validation combining **off-chain state simulation via Chainlink CRE** with **on-chain protocol guard contracts** for invariant and risk analysis.
- Implemented transaction decoding, NFT-based risk reporting, and execute/abort decision workflows; deployed across **Base, Ethereum, and Arbitrum testnets**.

### SettleOne — Yield-Generating Payment Commitment Protocol

[GitHub Repository](#)

*Solidity · Foundry · Chainlink · React · NodeJs · ExpressJs · Viem · DeFi Yield Strategies · StableCoins*

- Built a full-stack **payment commitment protocol** addressing delayed MSME/B2B payments through escrow-backed on-chain deal enforcement.
- Designed Solidity contracts for **deal lifecycle management, dispute arbitration, delivery verification, settlement, and fund release**.
- Integrated **yield-generating stablecoin escrow strategies** to improve capital efficiency during locked commitment periods; deployed across multiple testnets.

## DexGateway — Multi-Chain DEX Routing Infrastructure

[GitHub Repository](#)

Solidity · Foundry · React · NodeJs · ExpressJs · TypeScript · The Graph

- Built full-stack DEX routing infrastructure integrating **7 Uniswap V2-style DEXes across 9 blockchain networks**.
- Implemented unified routing logic for **token swaps, liquidity provisioning, and liquidity withdrawal** through router abstraction.
- Engineered liquidity discovery, execution routing, and frontend transaction workflows across fragmented liquidity environments.

## Protocol Invariant Checker — DeFi Protocol Security Framework

[GitHub Repository](#)

Rust · CLI · Security Tooling · Protocol Analysis

- Built a modular Rust framework for **automated DeFi protocol security analysis, invariant testing, and structured audit reporting**.
- Designed a protocol-agnostic architecture using a shared Protocol interface, enabling seamless integration of new protocol implementations.
- Generated standardized security reports with **severity classification, vulnerability categorization, and remediation recommendations**.

## SMART CONTRACT SECURITY EXPERIENCE

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Independent Smart Contract Security Researcher

[github.com/Sourav-IIITBPL/audits](https://github.com/Sourav-IIITBPL/audits)

[Sherlock](#) | [Code4rena](#) | [Cantina](#)

July, 2025 – present

- Audited **20+ production DeFi protocols**, including AMMs, lending systems, vault architectures, staking protocols, and DEX infrastructure.
- Identified **10+ validated vulnerabilities (2 High, 3 Medium, 6 Low)** with reproducible Proof-of-Concept exploit demonstrations.
- Specialized in **protocol accounting analysis, invariant reasoning, reentrancy surfaces, exploit validation, and adversarial edge-case analysis**.

## EDUCATION

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Indian Institute of Information Technology (IIIT), Bhopal

*Class of 2027*

B.Tech — Electronics & Communication Engineering

CGPA: 7.57 / 10 | Expected Graduation: 2027

Maharishi Vidya Mandir, Mandla

Senior Secondary (Class XII) — **89.4%**

## ACHIEVEMENTS

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- **Cyfrin Updraft Certified** — Advanced Solidity development, smart contract security, protocol testing, and DeFi protocol engineering ([Certification Profile](#)).
- **LeetCode Knight (1868 Rating)** — Solved **500+ DSA problems** with strong algorithmic problem-solving skills.
- **HackVision 2026** — **Top 100 finalist team** among **400+ teams** in a national virtual hackathon.
- **Smart India Hackathon 2024** — National-level participant and team lead.
- **AlgoUniversity Tech Fellowship** — Selected for **2024 & 2025 cohorts**.
- **HER DAO Rust Cohort 1 Graduate**.